

Predictive Modeling of Cattle Metabolic Disease Risk Using Bayesian Logistic Regression

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Data Science Thesis | Warren Wilson College



Research Problems

Goal:

- These diseases strike suddenly, and traditional diagnostics rely on thresholds or visual symptoms.
- My research addresses the gap by modeling individualized risk probabilities

Symptoms:

- Grass Tetany: A magnesium deficiency disorder in cattle causing muscle tremors, excitability, and sudden collapse, often triggered by grazing lush spring pastures.
- Milk Fever: A calcium deficiency around calving that leads to muscle weakness, cold extremities, and recumbency, especially in older dairy cows.

Traditional

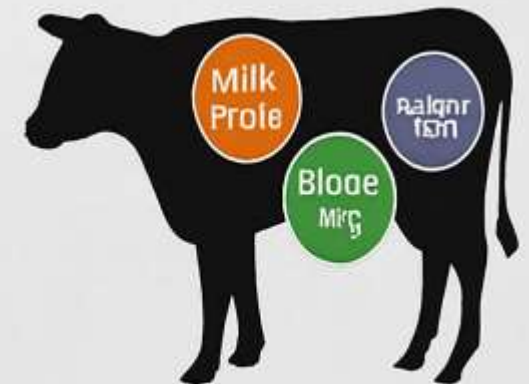


- Sudden onset, limited treatment window
- Misses individual variation

Predictive

Milk Fever →
hypocalcemiae (low calcium)

Grass Tetany →
hypomagnesemia or K Mg
imbalance



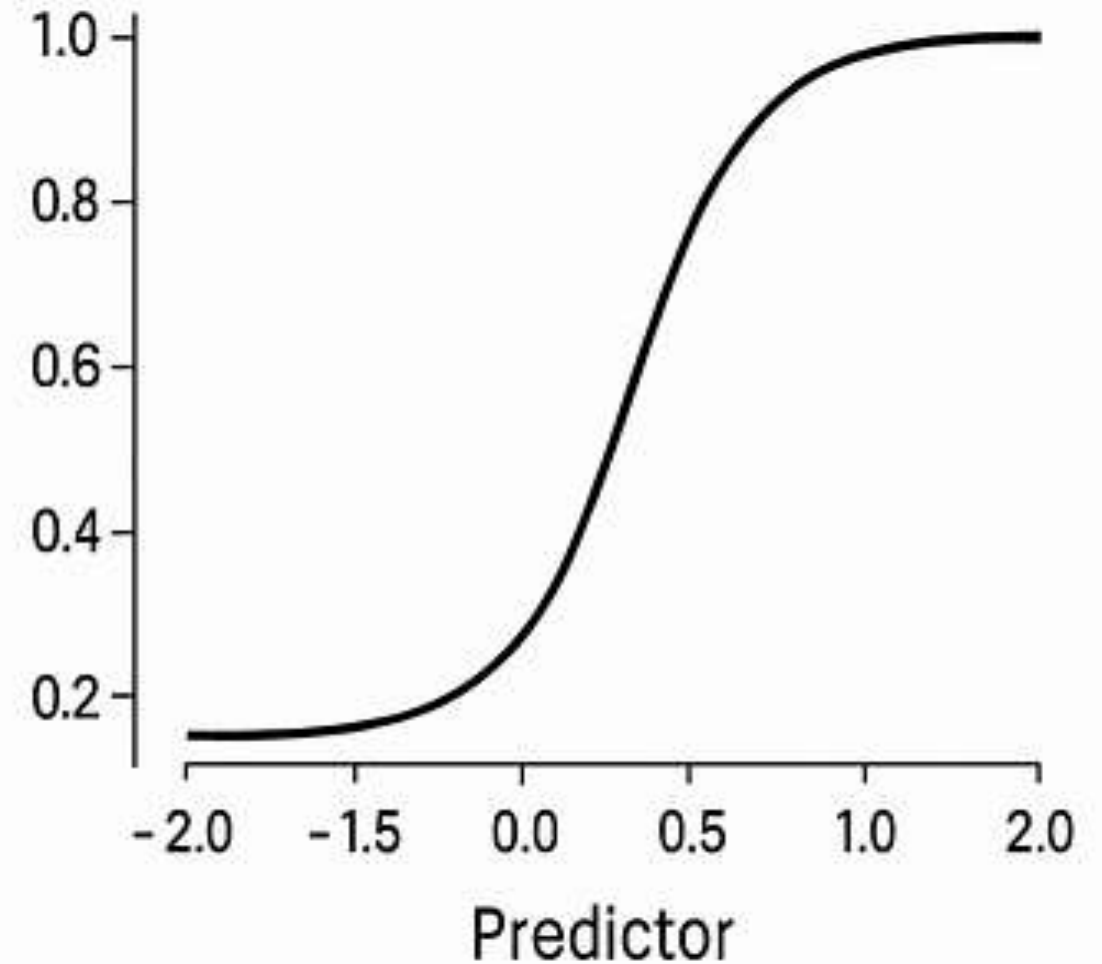
Objective & Data Sources

- **Objective:**
To develop interpretable machine learning models that detect early metabolic vulnerability by integrating mineral thresholds, behavioral indicators, and temporal dynamics.
- **Data Source:**
We use longitudinal cattle health data from Warren Wilson College's working farm, including:
- **Biochemical assays** (ICP-OES, AAS) for calcium, magnesium, and potassium levels
- **Behavioral metrics** such as chute score and exit velocity
- **Real-world conditions** reflecting ecological variability and physiological stress



Methodology Overview

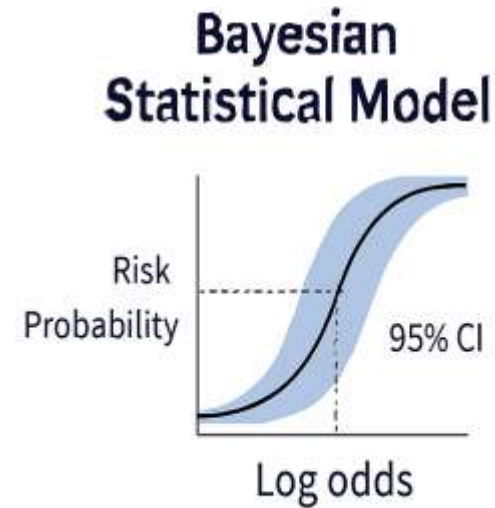
- **Model:** Bayesian logistic regression (rethinking::map)
- **Likelihood:** Bernoulli - $L(p) = \text{dbinom}(1, p) = p^1 (1 - p)^0$ defines outcome probabilities
- **Link Function:** Logistic - $\text{logit}(p) = \alpha + \sum_j \beta_j X_j$
transforms predictors into interpretable risk estimates
- **Predictors:** Standardized calcium, magnesium, potassium, chute score, exit velocity, encoded hair whorl
- **Labels:** Synthetic, physiologically grounded rules:
 - $\text{Mg} < 1.6$
 - $\text{K:Mg} > 3.0$
 - $\text{Ca} < 8.0$
- **Validation:** 4-fold cross-validation (folds: 10, 10, 10, 9) with fold-specific GT and MF models



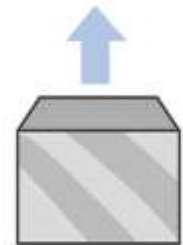
Why Bayesian (and not black-box ML?)

- **Probabilities, not flags:**
Outputs are continuous risk probabilities per cow, not binary yes/no labels.
- **Uncertainty is explicit :**
Posterior distributions and HPDIs quantify confidence, not just point estimates.
- **Interpretability is physiological:**
Coefficients map directly to mineral thresholds and behavioral indicators.
- **Data reality is respected:**
Bayesian methods handle small, imbalanced datasets without sacrificing rigor.
- **From Physiology to Probabilities:**
Bayesian models deliver interpretable, probabilistic insights grounded in physiology — not opaque predictions.”

Bayesian models deliver interpretable, probabilistic insights grounded in physiology — not opaque predictions.



Black-box Machine Learning



Risk
Yes / No?

Diagnostic Metrics for Model Evaluation

Sensitivity (True Positive Rate)

Formula:

$$\text{Sensitivity} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}}$$

Interpretation:

Sensitivity measures how well the model identifies sick cows. It answers:

“Out of all the cows that are actually sick, how many did the model correctly flag?” A sensitivity of 1 means the model flagged every sick cow correctly.

Specificity (True Negative Rate)

Formula:

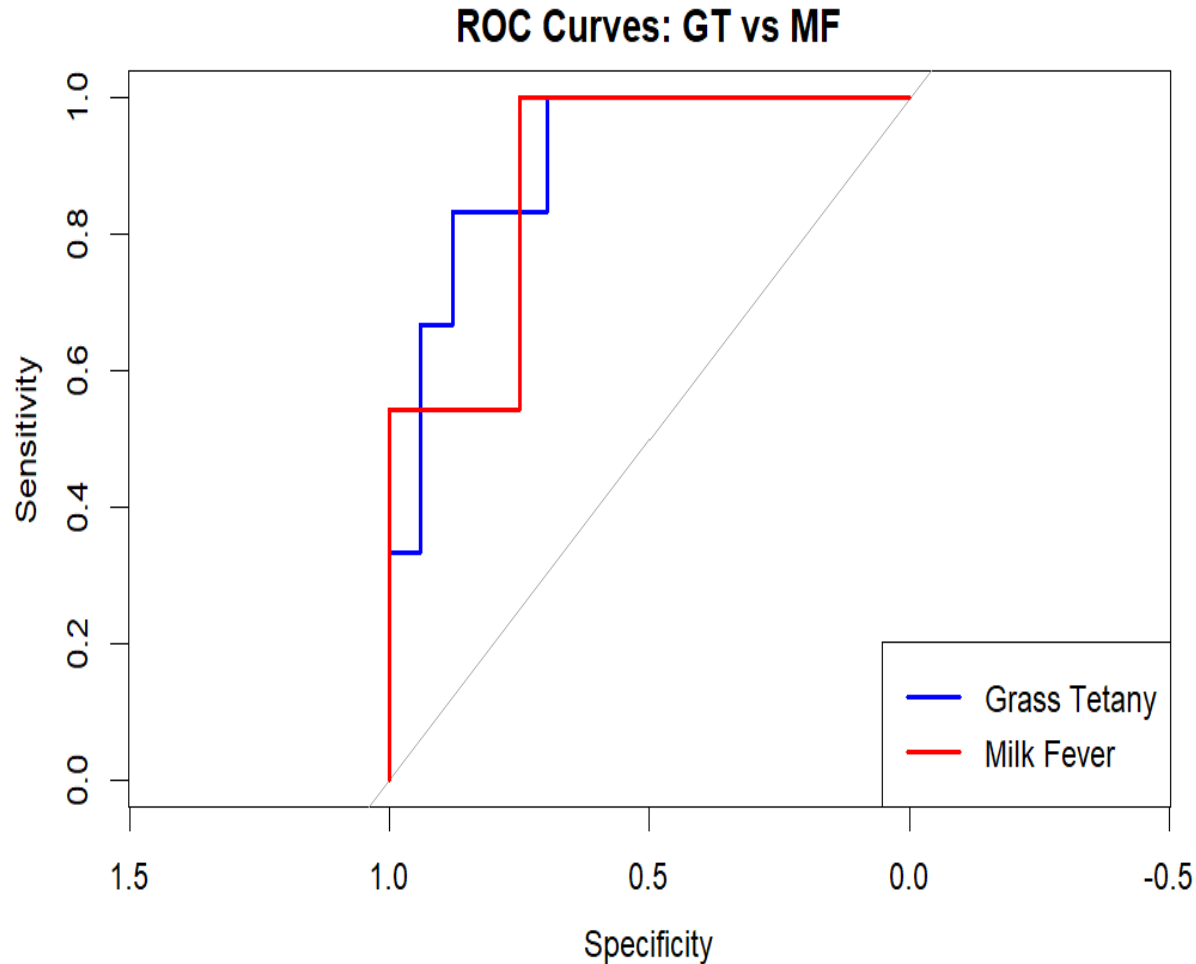
$$\text{Specificity} = \frac{\text{True Negatives (TN)}}{\text{True Negatives (TN)} + \text{False Positives (FP)}}$$

Interpretation:

Specificity measures how well the model identifies healthy cows. It answers:

“Out of all the cows that are actually healthy, how many did the model correctly leave unflagged?” A specificity of 1 means the model avoided false alarms and correctly identified all healthy cows.

ROC Curves – Sensitivity vs Specificity



- **Sensitivity (True Positive Rate):** Measures how well the model identifies sick cows. A sensitivity of 1 means the model flags every sick cow correctly.

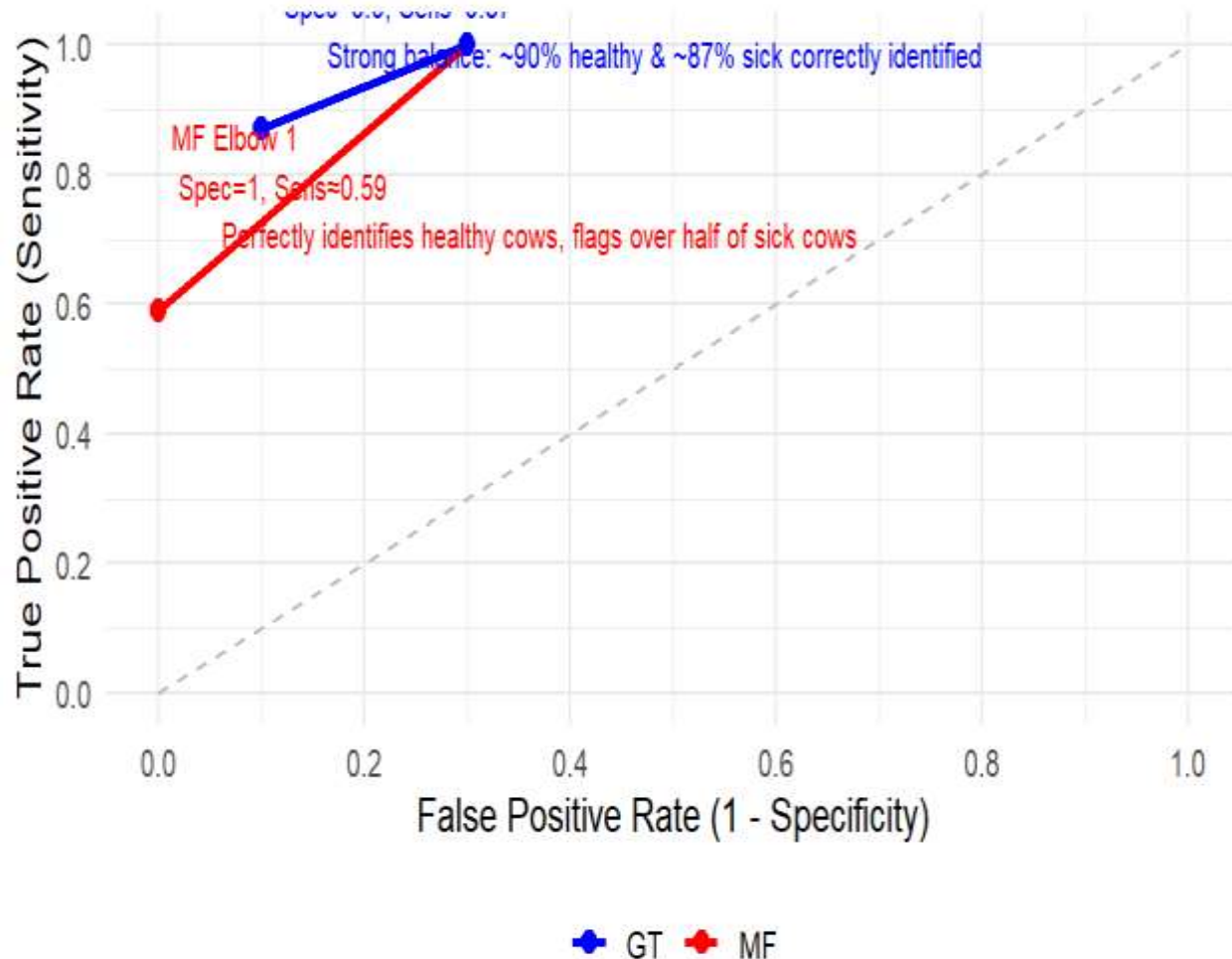
- **Specificity (True Negative Rate):** Measures how well the model identifies healthy cows. A specificity of 1 means the model avoids false alarms and correctly identifies all healthy cows.

- **False Positive Rate (1 - Specificity):** The proportion of healthy cows incorrectly flagged as sick.

- **Elbow Points:** Represent optimal trade-offs between sensitivity and specificity — where the model performs best diagnostically.

ROC Curve: Diagnostic Accuracy of Milk Fever vs Grass Tetany

ROC Curve: Diagnostic Performance for GT and MF



MF Elbow 1 (Red, Spec = 1, Sens ≈ 0.59): Perfectly identifies healthy cows, but misses ~40% of sick cows.

MF Elbow 2 (Red, Spec ≈ 0.7, Sens = 1): Flags all sick cows, correctly identifies 70% of healthy cows.

GT Elbow 1 (Blue, Spec ≈ 0.9, Sens ≈ 0.87): Strong balance ~90% healthy and ~87% sick cows correctly identified.

GT Elbow 2 (Blue, Spec ≈ 0.7, Sens = 1): Flags all sick cows, correctly identifies 70% of healthy cows.

Random Classifier (Dashed line): Baseline performance, showing chance-level separation.

Results Overview

ROC Curve Performance

- **ROC:**
 - *Milk Fever (MF)* curve shows stronger discrimination — clear separation from random classifier
 - *Grass Tetany (GT)* curve weaker — reflects challenge of rare-event prediction

Accuracy:

- **Model-based predictions:**
 - GT accuracy = **84.6%**
 - MF accuracy = **89.7%**

- **Rule-based flags (thresholds):**

- GT accuracy = **97.4%**
 - MF accuracy = **100%**
- *Insight:* Deterministic flags are highly accurate but lack nuance. Model predictions add uncertainty quantification and better generalization.

Coefficient Effects (Posterior HDPI)

- **Lower Ca and Mg** → strong increase in predicted risk
- **Chute and Exit scores** → weak associations with stress physiology
- **Hair whorl position** → small but directional behavioral effect

Confusion Matrix: Diagnostic Performance for GT and MF

This table compares the cows flagged as ‘at risk’ by our model against their actual health outcomes. It shows how well the risk-flagging aligns with reality.”

Grass Tetany (GT)

	Sick	Healthy
Flagged	TP: 5 Sick cows correctly flagged, no false alarms	FP: 0 No healthy cows incorrectly flagged
Not Flagged	FN: 1 Sick cow missed	TN: 33 Healthy cows correctly identified

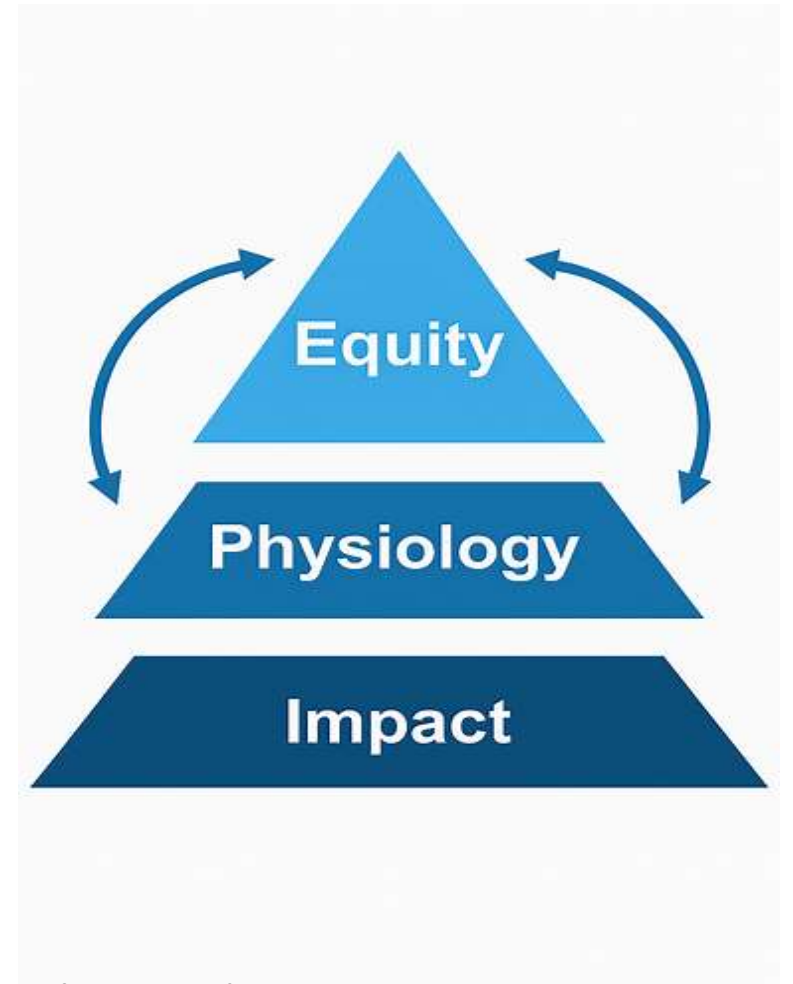
Milk Fever (MF)

	Sick	Healthy
Flagged	TP: 35 Sick cows correctly flagged, no false alarms	FP: 4 Healthy cows incorrectly flagged
Not Flagged	FN: 0 No sick cows missed (perfect sensitivity)	TN: 0 No healthy cows correctly identified

This confusion matrix shows that Grass Tetany detection achieved high specificity with one false negative, while Milk Fever detection reached perfect sensitivity but produced four false positives.

Conclusion – Key Narrative

- **Bayesian modeling strengthens herd health diagnostics** by combining accuracy, interpretability, and uncertainty quantification.
- **Milk Fever vs. Grass Tetany:** reliable, physiologically grounded predictions for Milk Fever; rare-event challenges but validated signals for Grass Tetany.
- **Broader impact:** actionable dashboards connect predictive analytics to real-world decision-making and translational health applications.



Predictive modeling is not just a statistical tool — it's a pathway to equity, physiology, and impact.

Citations

- *Xiao, S., Dhand, N. K., Wang, Z., Hu, K., Thomson, P. C., House, J. K., & Khatkar, M. S. (2025). Review of applications of deep learning in veterinary diagnostics and animal health. Frontiers in Veterinary Science, 12, 1511522. <https://doi.org/10.3389/fvets.2025.1511522> → Discusses how machine learning and deep learning are applied in veterinary diagnostics, including biomarker-based disease prediction.*
- *Stewart, A. J. (2025). Hypomagnesemic tetany in cattle and sheep. In MSD Veterinary Manual. Retrieved from <https://www.msdsvetmanual.com/metabolic-disorders/disorders-of-magnesium-metabolism/hypomagnesemic-tetany-in-cattle-and-sheep> → Provides veterinary background on Grass Tetany, its link to magnesium deficiency, and diagnostic thresholds.*
- *Harris, S., Shallcrass, M., & Cohen, S. (2024). A review of animal-based welfare indicators for calves and cattle. Ruminants, 4(4), 565–601. <https://doi.org/10.3390/ruminants4040040> → Reviews behavioral indicators (chute and exit scores, hair whorl) as predictors of cattle stress and health.*
- *Hitchcock, D. B. (2024). Bayesian logistic regression models. STAT 535: Chapter 13 Lecture Notes. University of South Carolina. Retrieved from <https://people.stat.sc.edu/hitchcock/stat535slides13BRBhandout.pdf> → Explains Bayesian logistic regression foundations, including dbinom and logit functions used in my modeling.*
- *Lee, S. (2025). Precision–Recall Curve: A comprehensive guide. Number Analytics Blog. Retrieved from <https://www.numberanalytics.com/blog/precision-recall-curve-comprehensive-guide> → Explains PR curves and why they are critical for imbalanced datasets like cattle disease prediction.*

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Thank you